

ENV 4120

Air Pollution and Hazardous Waste Control

Today's Topics

- Introduction to air pollution
- Expectation
 - Understand Tragedy of the Commons
 - Familiar with air pollution issues at different scales
 - Understand CAA, NAAQS, CAP, HAP, NESHAPS, SIP, NSR, PSD

The Tragedy of the Commons

by Garrett Hardin

Science, Dec. 13, 1968

vol. 162, pp. 1243-1248

- **Tragedy for cattle herdsman (Lloyd, 1833)**
 - 10 herdsman each with 100 cattle; at sustainable limit of commons; if herdsman adds 1
 - Increase individual benefit, but decrease shared resources
 - Rational decision of each herdsman: keep adding animals
 - Result: commons overgrazed and destroyed
- **Growing pistachio trees in CA**
 - Pistachio needs lots of water
 - Drought → pistachio price went up → grow more pistachio → drought more severe

Air pollution issues at different scales

What is air pollution

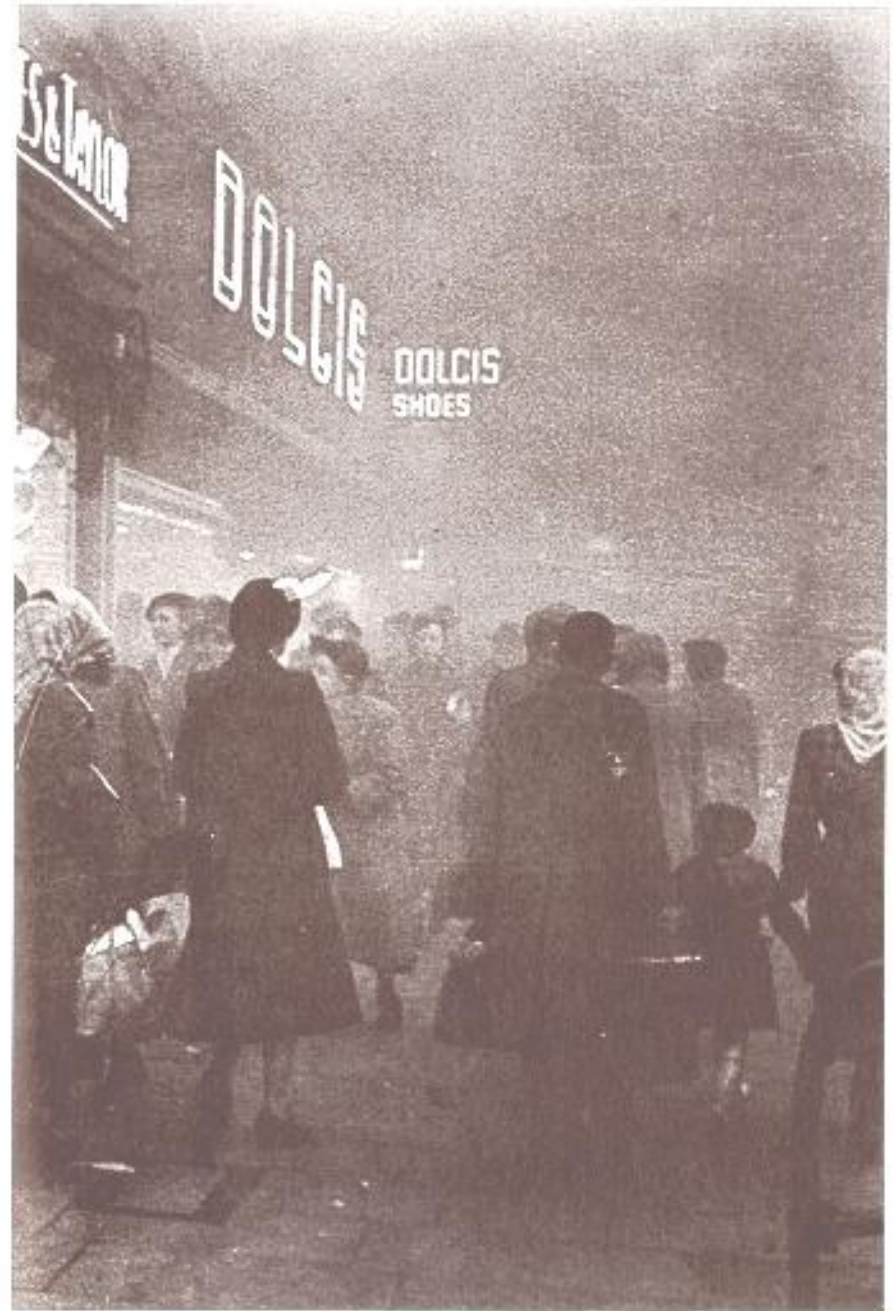
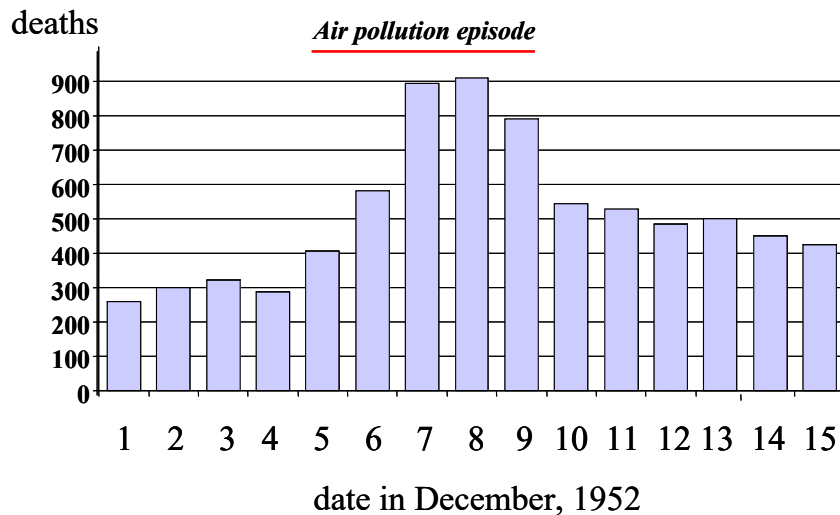
- As defined in Florida Administrative Code:
- “Air pollution is the presence in the outside atmosphere ... of any one or more substances or pollutants in quantities which are or may be harmful or injurious to human health or welfare, animal or plant life, or property, or unreasonably interfere with the enjoyment of life or property, including outdoor recreation.”

Scales of air pollution problems

- Personal
- Micro-environmental (meters) – indoor air pollution
- Local (< 5 km) – hot spots, near source
- Urban (~ 50 km) – urban smog
- Regional (50 – 500 km) – acid deposition, visibility
- Continental (500 – 1000 km) – long-range transport, stratospheric ozone depletion
- Global – stratospheric ozone depletion, long range transport, climate change

London smog, 1952

- December temperature inversion
- air pollution levels (PM, SO₂) 5x higher than normal
- 4,000 excess deaths attributed to fog
- 80% of deaths due to cardiorespiratory disease (30% typ)



LA smog



Beijing smog



Smog in 20 minutes. Beijing, China. Jan 2nd, 2017
Credit: YouTube/Chas Pope

Regional air pollution (haze)

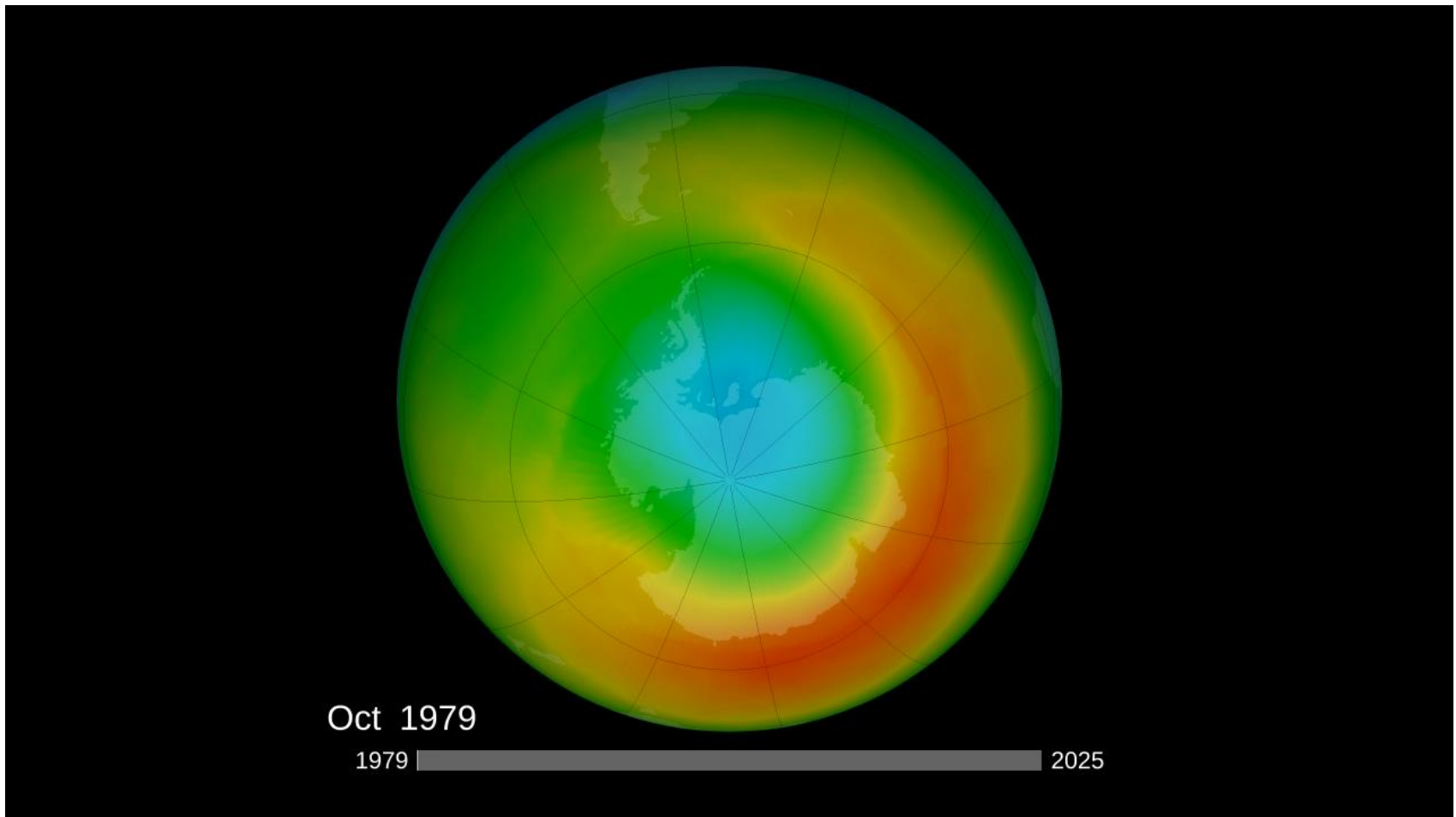
Shenandoah Valley



Acid deposition

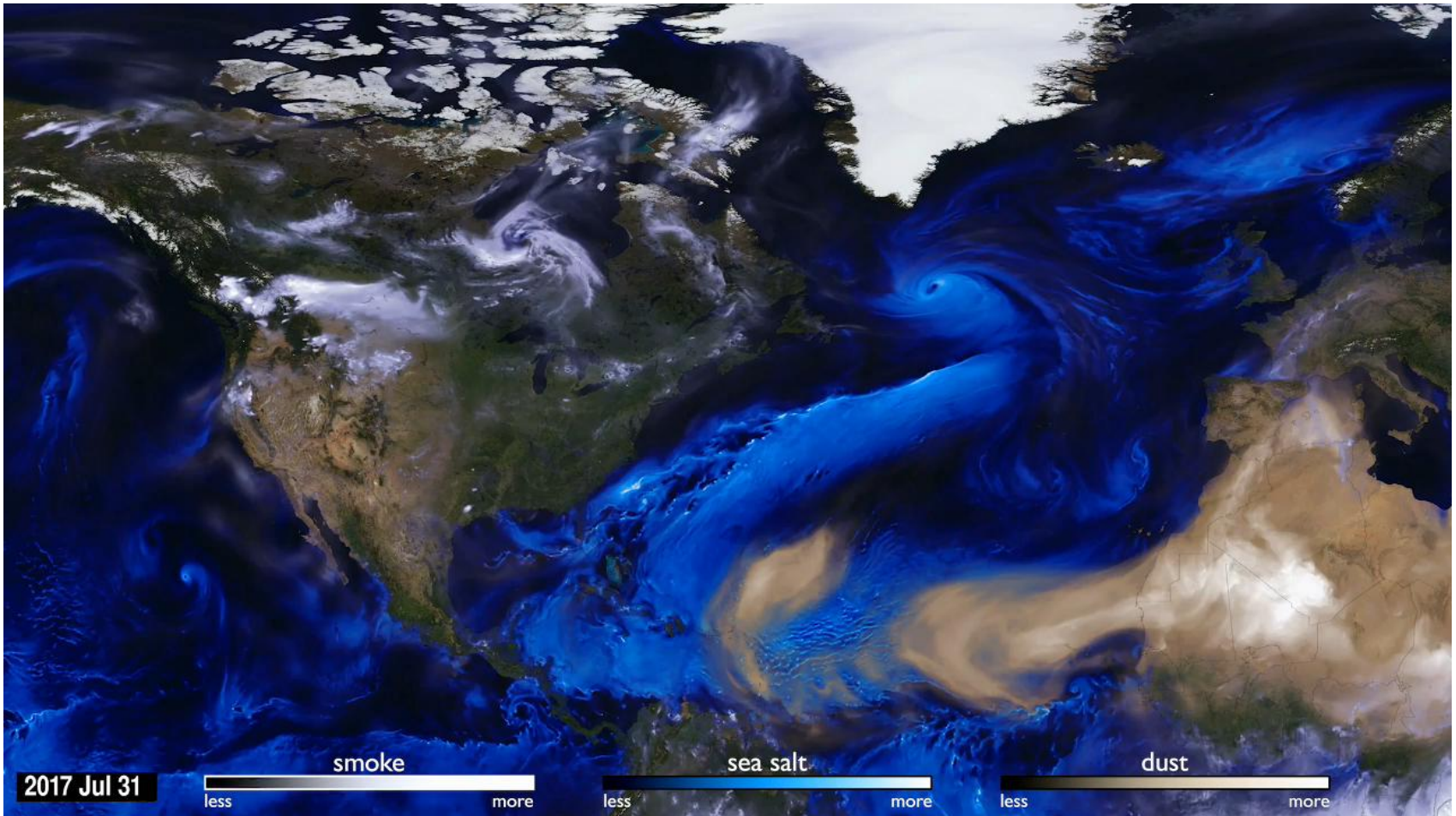


Stratospheric ozone depletion



Antarctic September ozone hole from 1979 to 2022
(NASA Ozone watch <http://ozonewatch.gsfc.nasa.gov/monthly/SH.html>)

Wildfires



2017 Hurricanes and Aerosols Simulation from NASA
(<https://svs.gsfc.nasa.gov/12772>)
Latest satellite image of fire see NASA worldview

Regulation

Brief U.S. Federal Legislative History

- Mid 1800s: industrialized cities were well established as very dirty places
 - New York, Chicago, Pittsburgh etc...



Pittsburgh from the Salt Works at Saw Mill, by William T. Russell, 1843
(<http://explorepahistory.com/displayimage.php?imgId=1-2-D44>)

Brief U.S. Federal Legislative History

- By turn of the century, air pollution becoming recognized as a local health problem
- 1907: The International Association for the Prevention of Smoke established
 - Later the Air Pollution Control Association, now the Air & Waste Management Association
- Automobile revolution and beginnings of American consumer culture

Brief U.S. Federal Legislative History

- 1940s: LA haze; Donora (PA) smog (20 deaths)
- 1952: London smog (4000 deaths)
- 1955: Clean Air Legislation (1st Fed. regulation)
 - Authorized Public Health Service (Dept. of Health, Education, and Welfare: DHEW) to:
 - Conduct air pollution research and training programs
 - Provide technical assistance to state and local governments

London smog
(www.klimaforschung.net)



Donora, PA at noon on
October 29, 1948
<http://pabook.libraries.psu.edu/palitmap/DonoraSmog.html>

Brief U.S. Federal Legislative History

- 1960s: Social change and significant national environmental movements
 - Sierra Club, Wilderness Society
 - 1962: Rachel Carson's 'Silent Spring'
- 1963: Clean Air Act (CAA)
 - Broadened federal role in air quality management
 - Grants for program development/improvement of state and local air pollution control
 - Research, training, and technical assistance
 - Federal enforcement authority for interstate air pollution problems
 - Federal research responsibility for automobile and SO_x pollution
 - Authority to compile and publish criteria for pollutants

Brief U.S. Federal Legislative History

- 1965: Motor Vehicle Air Pollution Control Act
 - Amendment to 1963 CAA
 - Established the National Air Pollution Control Administration (NAPCA)
 - Authorized DHEW/NAPCA to promulgate motor vehicle emissions standards

Brief U.S. Federal Legislative History

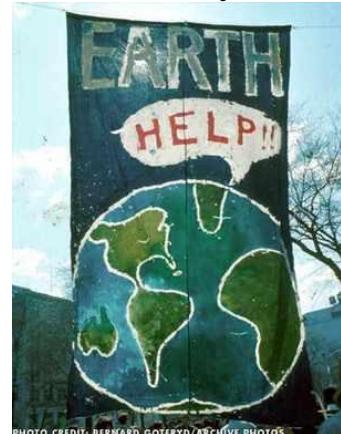
- 1967: Air Quality Act (Amendment to CAA)
 - Institution of a regional approach to air pollution control and standard setting
 - Air Quality Control Regions (AQCR)
 - NAPCA to develop/issue air quality criteria and control technique info
 - States to develop state air quality standards, **state implementation plan (SIP)**



Brief U.S. Federal Legislative History

- 1970: USEPA established (NAPCA eliminated)
- 1970: Clean Air Act Amendments
 - Provided for federal enforcement authority
 - Stringent auto emissions and fuel standards
 - National Ambient Air Quality Standards (NAAQS)
 - Six Criteria Air Pollutants (CAP)
 - New Source Performance Standard (NSPS)
 - National Emissions Standards for Hazardous Air Pollutants (NESHAPS)

First earth day, 1970



Brief U.S. Federal Legislative History

- 1977: Clean Air Act Amendments
 - Required review of NAAQS every 5 years
 - Non-compliance penalties
 - New Source Review (NSR) for major sources
 - Prevention of Significant Deterioration (PSD)
 - Grant USEPA authority to regulate ozone destroying chemicals
- 1977: National Atmospheric Deposition Program (NADP) established
 - Monitoring trends of acidity in precipitation

Brief U.S. Federal Legislative History

- 1978: US ban manufacture and sale of CFCs for spray cans
- 1980: Acid Precipitation Act
 - National Acid Precipitation Assessment Program (NAPA)
 - Enlarged NADP to National Trends Network (NTN)
- 1980: Election of Ronald Reagan
 - “Onslaught” on environmental regulations

Brief U.S. Federal Legislative History

- 1990: Clean Air Act Amendments
 - Reauthorization and update of CAA
 - Strengthen measures for attaining air quality standards
 - Boundaries of non-attainment areas to include all areas in metropolitan area
 - Expanded regulation of HAPs
 - Max achievable control technology (MACT)
 - Establish program for prevention of accidental releases
 - Updated authority and specific requirements for stratospheric ozone depletion
 - Expanded enforcement powers and penalties
 - Criminal sanctions for negligent violations

Brief U.S. Federal Legislative History

- 2005: Clean Air Interstate Rule (CAIR)
 - Cap and trade for O₃ and PM
 - Replaced by Cross-State Air Pollution Rule (CSAPR) in 2011
- 2007: Supreme court ruling providing EPA authority over CO₂
- 2015: Volkswagen emissions scandal (\$2.8B fine)
- 2015: Clean Power Plan: reduce carbon emissions from power plants
- 2016: 1st Trump presidency
 - Many environmental policy changes followed

Brief U.S. Federal Legislative History

- 2017: Dismantle of Clean Power Plan
 - Held by supreme court in 2016
 - Replaced with Affordable Clean Energy rule in 2019
 - One example of Clean Coal: Kemper Project
- 2017 & Beyond: ?????
 - Pull out of Paris climate accord, twice
 - Inflation Reduction Act
 - Drill, baby, drill
 - Biggest USEPA deregulatory action in U.S. History
 - & countless others